3.3W Mono Class-D Audio Power Amplifier with Non-Clip and Power Limit

DESCRIPTION

The EUA2068A is a high efficiency, 3.3W mono class-D audio power amplifier with Non-Clip and Power Limit. A low noise, filterless PWM architecture eliminates the output filter, reducing external component count, system cost, and simplifying design.

EUA2068A having power limit interlock non-clip function, controls gain automatically and prevents output signals from being clipped at the supply voltage. In addition, this power limit function permits the maximum output to be arbitrarily specified to 100mW to 1000mW according to a speaker to use.

Furthermore, EUA2068A also integrate overcurrent and overtemperature protection functions, DC output protection function, under-voltage malfunction prevention function, as well as EMI noise and the pop noise reduction functions.

The EUA2068A is available in space-saving WCSP package.

FEATURES

- Unique Modulation Scheme Reduces EMI Emissions
- Unique Non-Clip function , Variable NCNA, NCNB,NCNOFF mode Select
- 3.3W 4-Ω Load at V_{DD}=5V, THD+N <10%
- Low Quiescent Current, I_{DD}=4mA @5V
- 2.5V to 5.5V Wide Supply Voltage
- Overcurrent Protection Function (OCP)
- Overtemperature Protection Function (OTP)
- DC Output Protection Function (DCOP)
- Under-voltage Malfunction Prevention Function (UVL)
- Short Circuit Auto-Recovery
- Power Limit Function
- 1.5mm × 1.5mm Wafer Chip Scale Package (WCSP)
- RoHS Compliant and 100% Lead(Pb)-Free Halogen-Free

APPLICATIONS

- Smart Phones/Cellular Phones
- Portable Multimedia Players

Typical Application Circuit

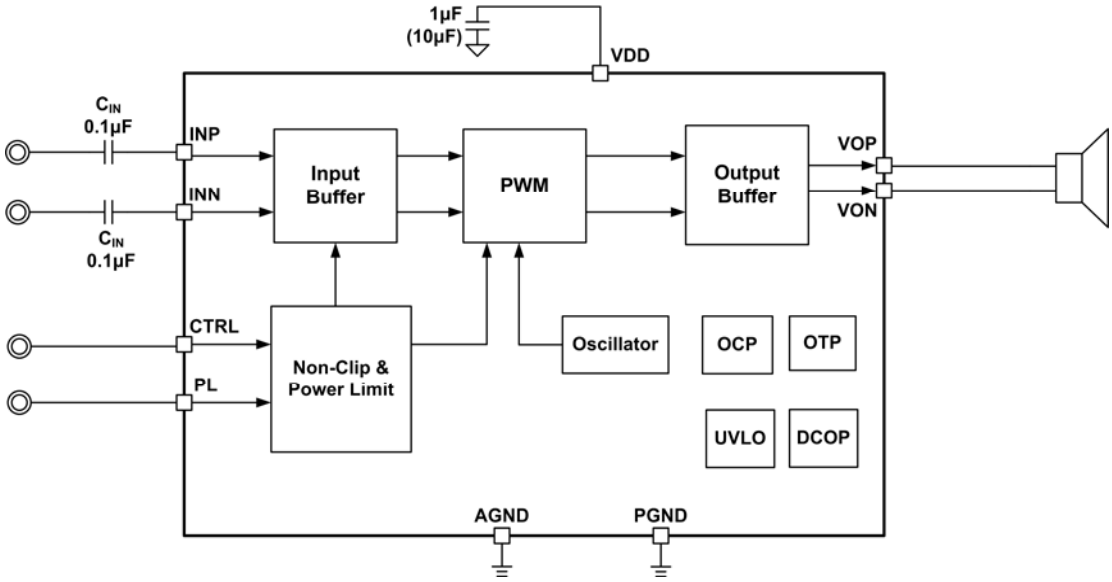
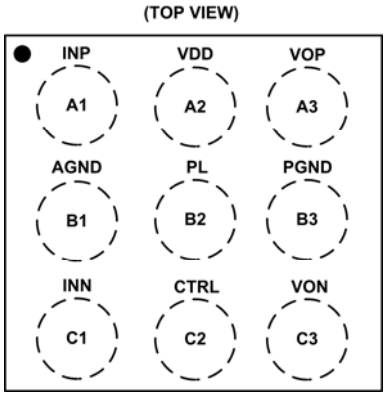


Figure1.

Pin Configurations

Package Type	Pin Configurations
WCSP-9	

Pin Description

PIN	WCSP-9	I/O	DESCRIPTION
INP	A1	A	Positive input terminal (differential +)
VDD	A2	Power	Power supply
VOP	A3	O	Positive output terminal (differential +)
AGND	B1	GND	GND for analog circuits
PL	B2	A	Power limit setting pin
PGND	B3	GND	GND for output
INN	C1	A	Negative input terminal (differential -)
CTRL	C2	I	Shut down and Non-clip control terminal
VON	C3	O	Negative output terminal (differential -)

(Note) I: Input terminal O: Output terminal A: Analog terminal

Ordering Information

Order Number	Package Type	Marking	Quantity per Reel	Operating Temperature Range
EUA2068AHIR1	WCSP-9	xxx L00	3000	-40 °C to +105°C

EUA2068A □ □ □ □

- Lead Free Code
  1: Lead Free, Halogen-Free
- Packing
  R: Tape & Reel
- Operating temperature range
  I: Industry Standard
- Package Type
  H: WCSP

Absolute Maximum Ratings

- Supply Voltage, V_{DD} ---------- -0.3 V to 6V
- Input Pin Voltage, INP, INN ---------- V_{GND} -0.6V to V_{DD} +0.6V
- Input Pin Voltage, other input pins ---------- V_{GND} -0.3V to V_{DD} +0.3V
- Junction Temperature, T_{JMAX} ---------- 160°C
- Storage Temperature Rang, T_{stg} ---------- -65°C to 150°C
- Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds ---------- 260°C
- Thermal Resistance

  θ_{JA} (WCSP-9) ---------- 110°C/W
- ESD Susceptibility (HBM) ---------- 2kV

Recommended Operating Conditions

	Min.	Max.	Unit
Supply voltage, V_{DD}	2.5	5.5	V
Operating free-air temperature, T_A	-40	105	°C

Consumption Current

(V_{DD}=3.6V, V_{GND}=0V, T_A =25°C, CTRL=V_{MOD3}, PL=V_{DD}, R_L=8Ω, Unless otherwise noted)

Symbol	Parameter	Conditions	EUA2068A			Unit
			Min.	Typ.	Max.	
I_{DD}	Consumption current	No input, no load	-	3.2	-	mA
I_{SD}	Shut-down mode consumption current	CTRL=V_{MOD4}	-	0.5	-	μA

Note (1): In the measurement conditions with R_L=8Ω, 8Ω (pure resistor)+ 30μH is used.

DC Characteristics

(V_{DD}=2.5V to 5.5V, V_{GND}=0V, T_A =-40°C to 105°C, PL=V_{DD}, Unless otherwise noted)

Symbol	Parameter		Conditions	EUA2068A			Unit
				Min.	Typ.	Max.	
V_{UVLH}	Startup threshold voltage			-	2	-	V
V_{UVLL}	Startup threshold voltage			-	1.8	-	V
V_{MOD1}	CTRL control pin input threshold voltage	Non-Clip A		1.5	1.6	V_{DD}	V
V_{MOD2}		Non-Clip B		1.0	1.18	1.35	
V_{MOD3}		Non-Clip OFF		0.65	0.75	0.85	
V_{MOD4}		Shut-down		V_{GND}	-	0.1	
V_{PDIS}	PL pin input threshold voltage	Power limit OFF		0.43 × V_{DD}	-	V_{DD}	V
V_{PENA}		Power limit enable mode		0.45	-	0.37 × V_{DD}	
V_{PFIX}		Power limit fix mode		0	-	0.1	
V_{OTPH}	High temperature protection function activated temperature			-	160	-	°C
V_{OTPL}	High temperature protection function released temperature			-	125	-	°C
I_{CTRL}	Input leakage current		CTRL pin	-1	-	+1	μA

AC Characteristics

(V_{DD}=2.5V to 5.5V, V_{GND}=0V, T_A =-40°C to 105°C, CTRL=V_{MOD3}, PL=V_{DD}, Unless otherwise noted)

Symbol	Parameter		Conditions	EUA2068A			Unit
				Min.	Typ.	Max.	
T_{STUP}	Startup time (shut-down cancel)			-	18	-	msec
T_{PDWL}	Shut-down setting time			-	6	-	msec
T_{PDWH}	Active setting time			-	25	-	msec
V_{DCDET}	DC error detection voltage		V_{DD}=3.6V	-	100	-	mV
T_{DCDET}	DC error detection time			-	1.3	-	sec
fc	Input cut-off frequency		C_{IN}=1.0μF (2)	-	16	-	Hz
T_{ATA}	Non-Clip A	Attack time	CTRL=V_{MOD1}	-	0.39	-	msec /dB
T_{RLA}		Release time		-	330	-	
T_{ATB}	Non-Clip B	Attack time	CTRL=V_{MOD2}	-	0.08	-	msec /dB
T_{RLB}		Release time		-	330	-	

Note (2): C_{IN}: DC-cut capacitor at the input stage

Analog Characteristics

(V_{DD}=3.6V, V_{GND}=0V, T_A =25°C, CTRL=V_{MOD3}, PL=V_{DD}, A_V=18dB, R_L=8Ω, 1kHz, sine wave, Unless otherwise noted)

Symbol	Parameter	Conditions	EUA2068A			Unit
			Min.	Typ.	Max.	
P_O	Maximum output	V_{DD} =5.0V, R_L=4Ω,THD=10%	-	3.3	-	W
		V_{DD} =3.6V, R_L=8Ω,THD=10%	-	0.95	-	
		V_{DD} =4.2V, R_L=8Ω,THD=1%	-	1.00	-	
		V_{DD} =3.6V, R_L=8Ω,THD=1%	-	0.75	-	
P_{ONC}	Non-Clip maximum output	V_{DD} =5.0V, R_L=4Ω, CTRL=V_{MOD1}	-	2.6	-	W
		V_{DD} =3.6V, R_L=8Ω, CTRL=V_{MOD1}	-	0.70	-	
P_{OPL}	Power limit output	PL=0V (fix mode)	-	600	-	mW
A_V	Voltage gain		-	18	-	dB
V_O	Output offset voltage		-	±3	-	mV
η	Efficiency			91		%
PSRR	f=217Hz, Vripple=0.1Vpp			65		dB
SNR		V_{DD} =5.0V, R_L=4Ω,BW=20KHz, A filter		91		dB

Note (3): C_{IN}: DC-cut capacitor at the input stage

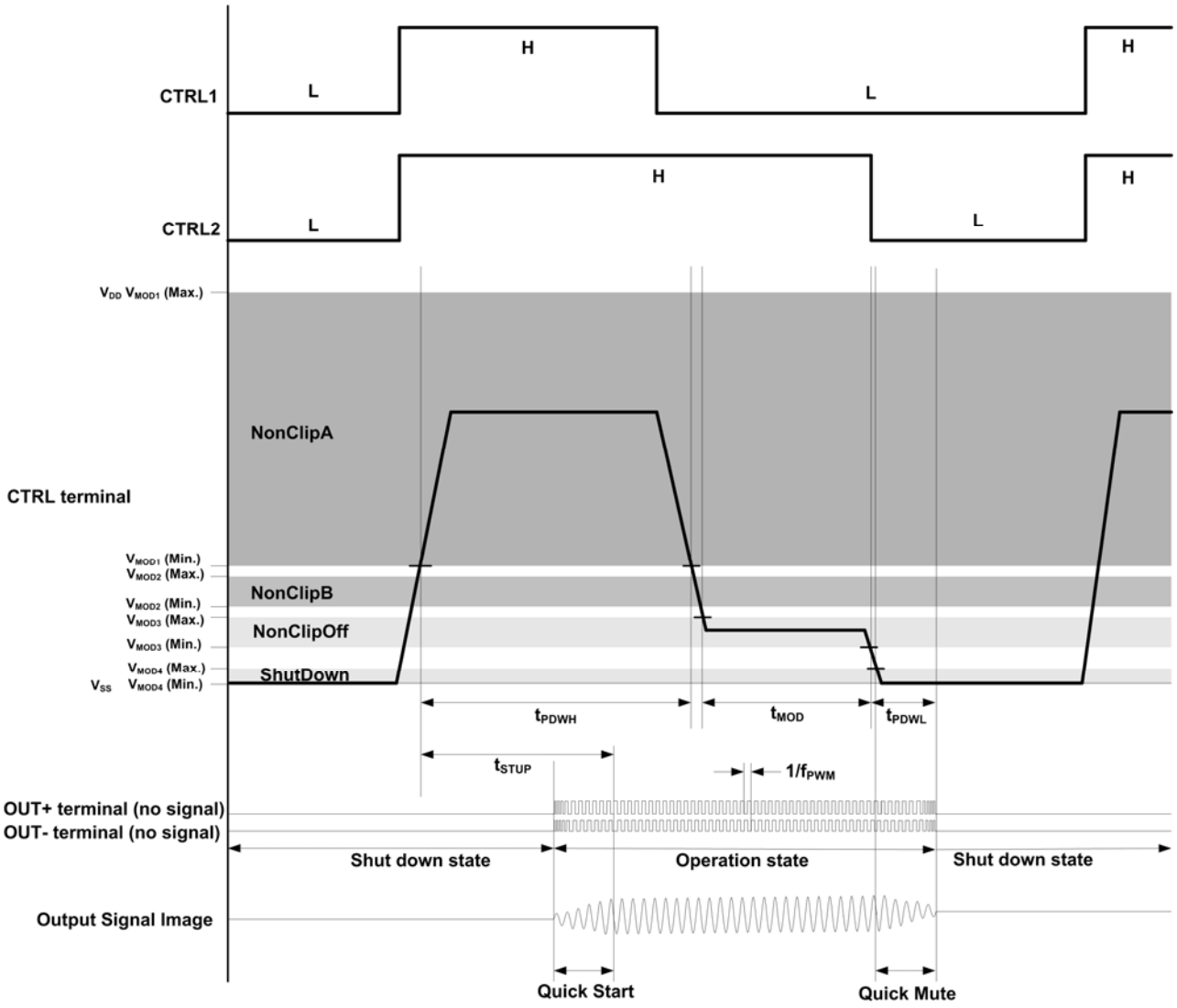

CTRL1
CTRL2
L
H
L
H
L
H
L
H
VDD VMOD1 (Max.)
NonClipA
CTRL terminal
VMOD1 (Min.)
VMOD2 (Max.)
NonClipB
VMOD2 (Min.)
VMOD3 (Max.)
NonClipOff
VMOD3 (Min.)
VMOD4 (Max.)
VSS VMOD4 (Min.)
ShutDown
tPDWH
tMOD
tPDWL
tSTUP
1/fPWM
OUT+ terminal (no signal)
OUT- terminal (no signal)
Shut down state
Operation state
Shut down state
Output Signal Image
Quick Start
Quick Mute

Figure2.

Typical Operating Characteristics

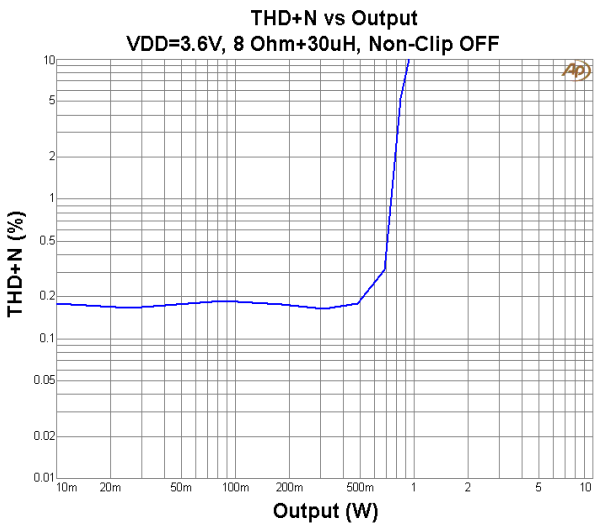


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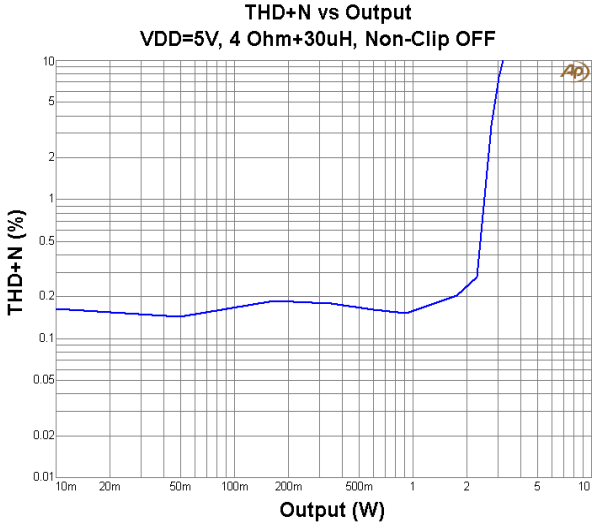


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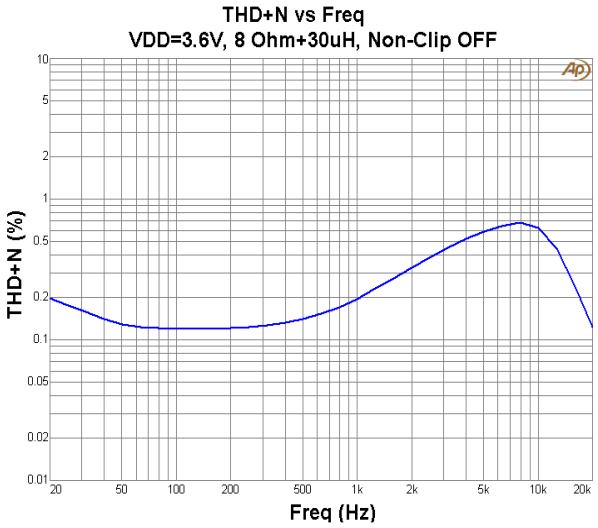


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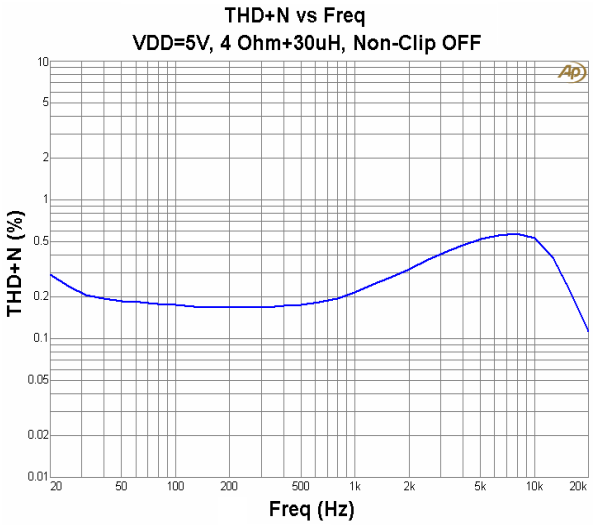


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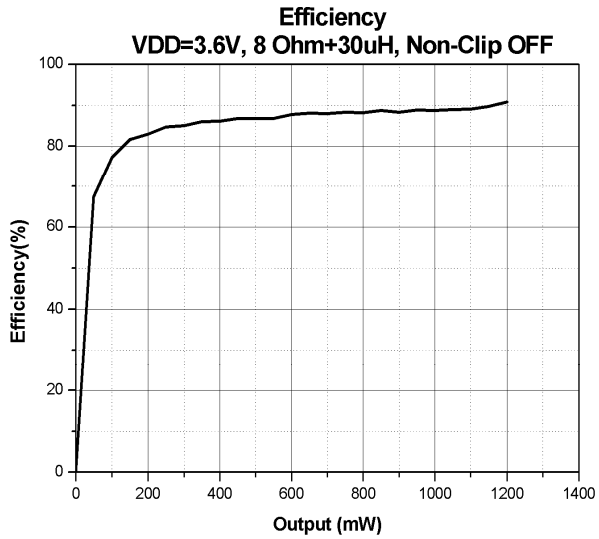


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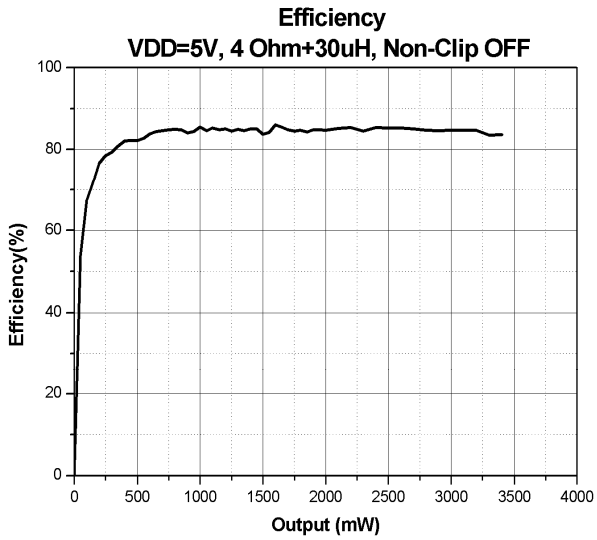


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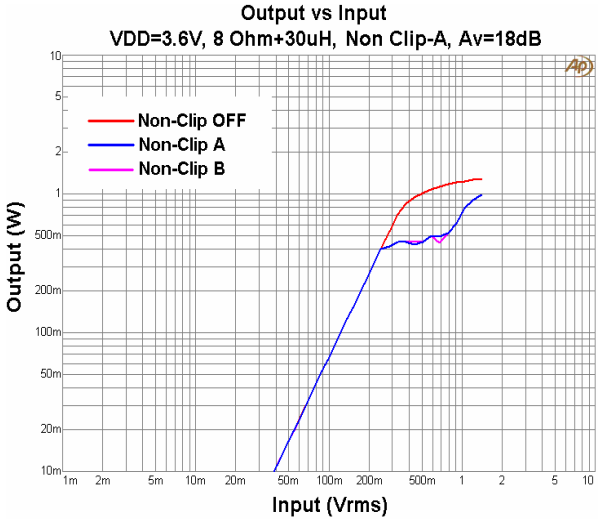


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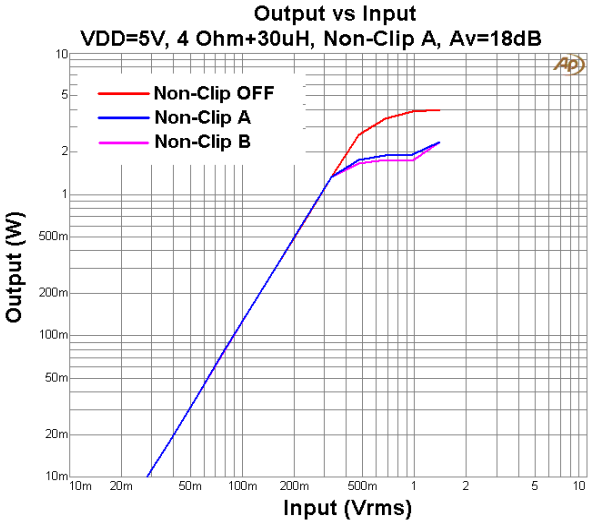


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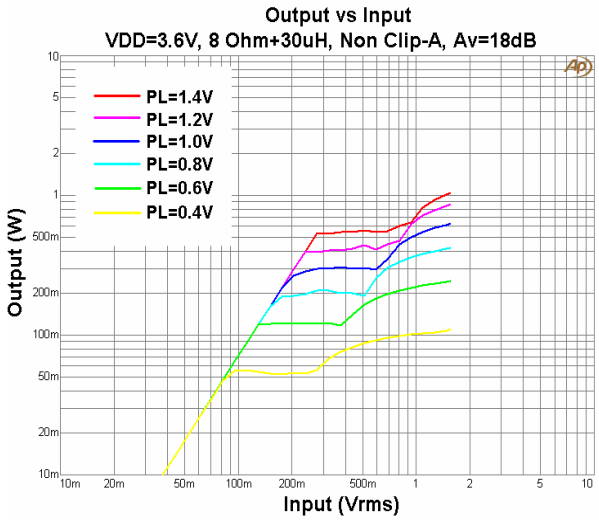


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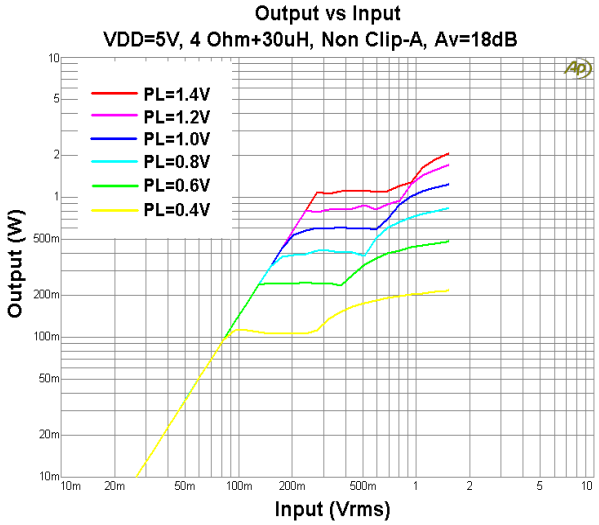


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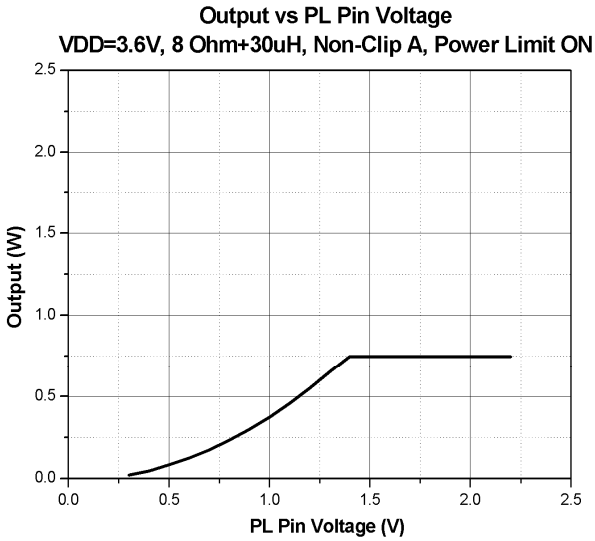


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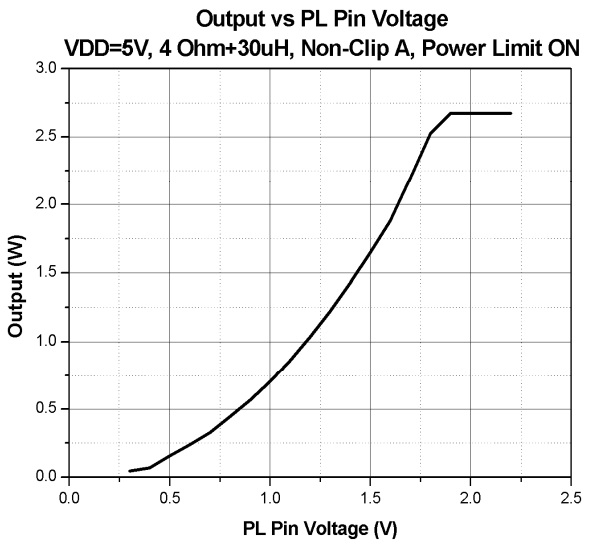


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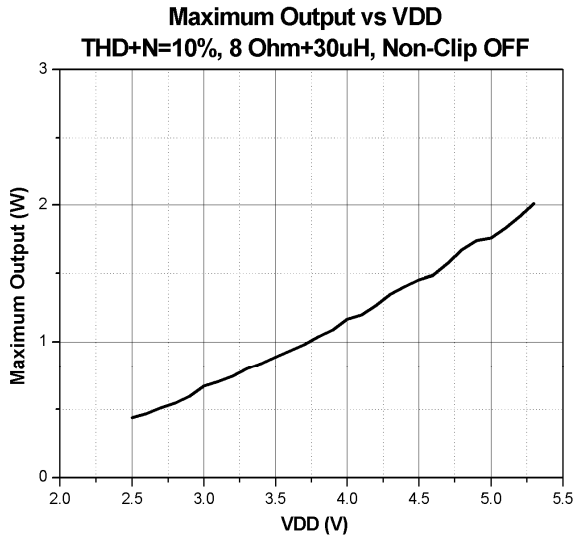


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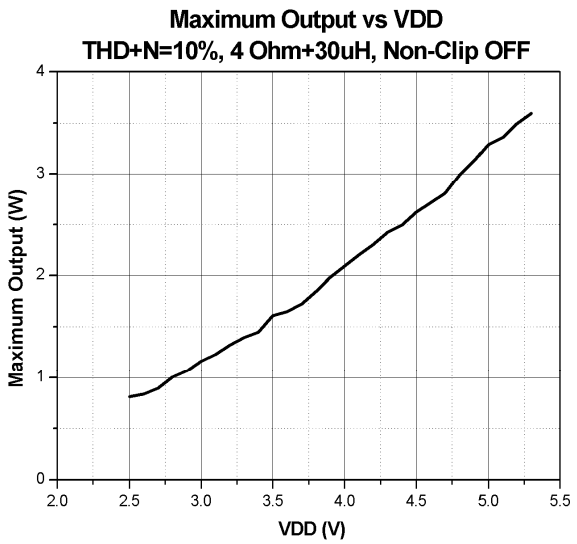


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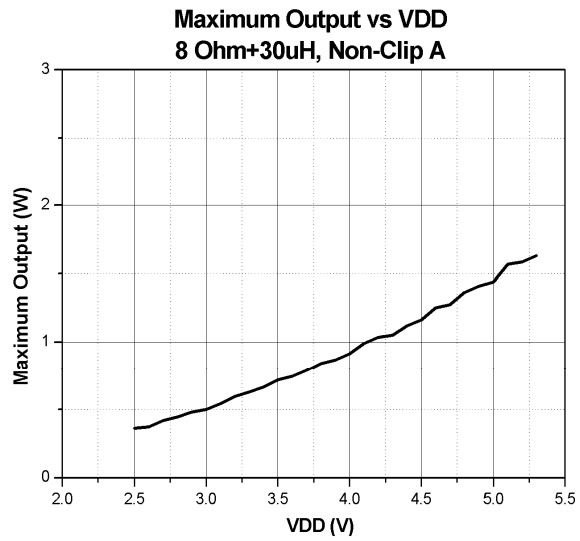


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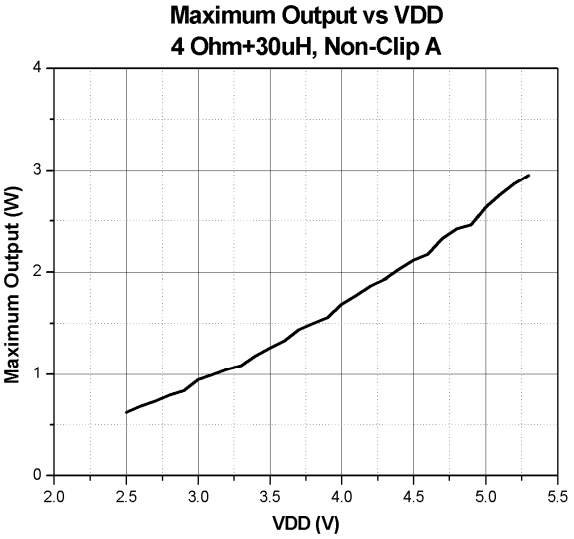


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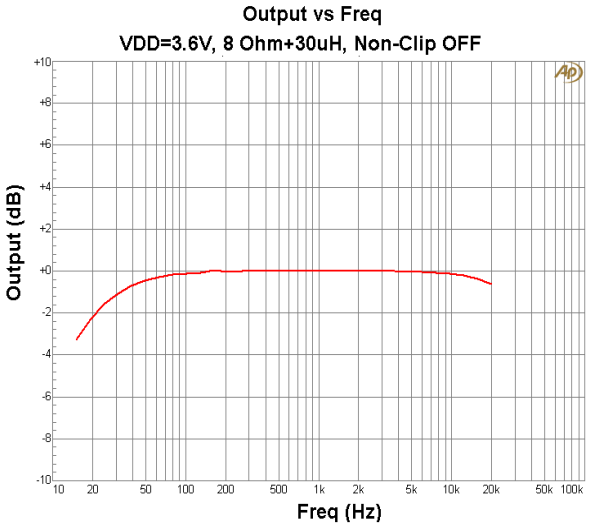


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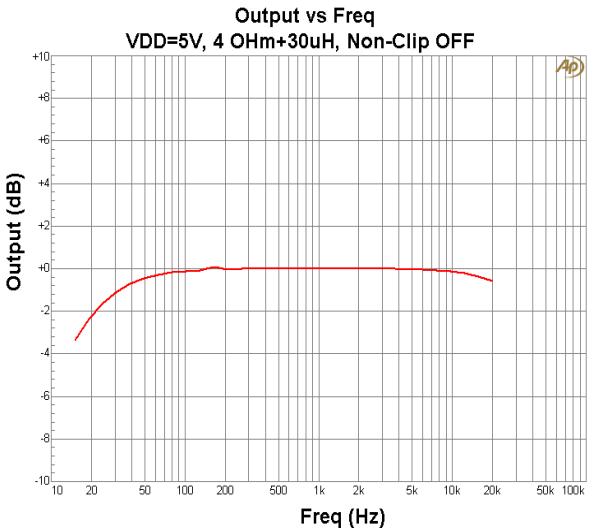


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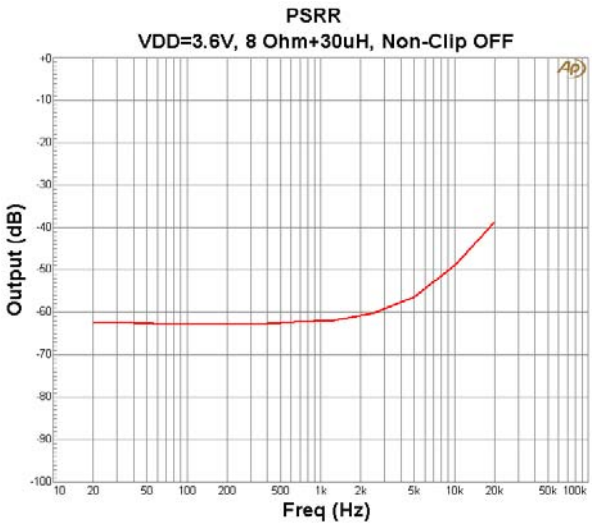


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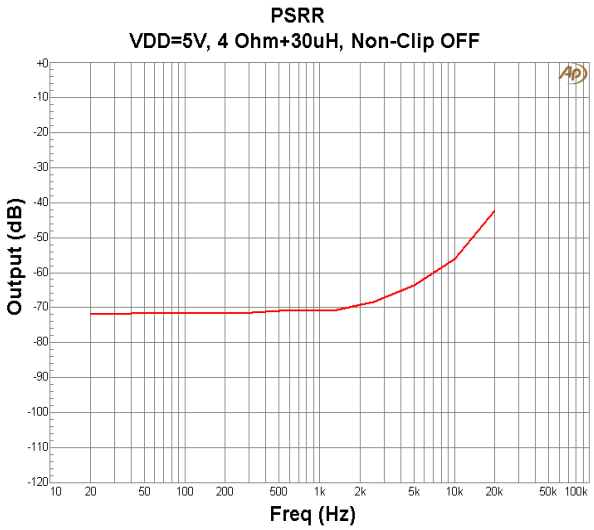


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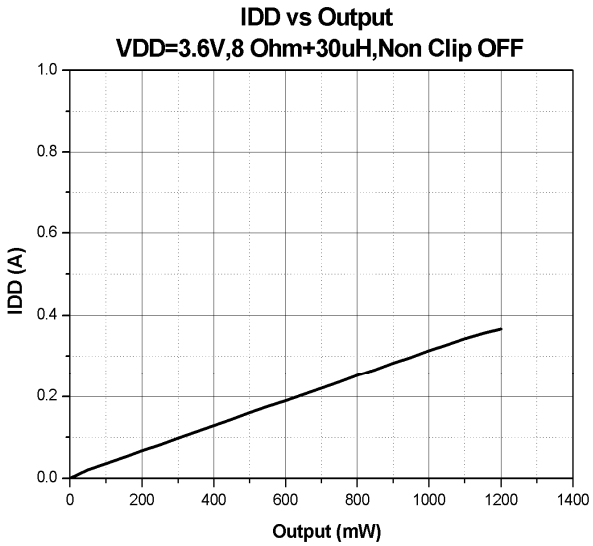


Figure23.

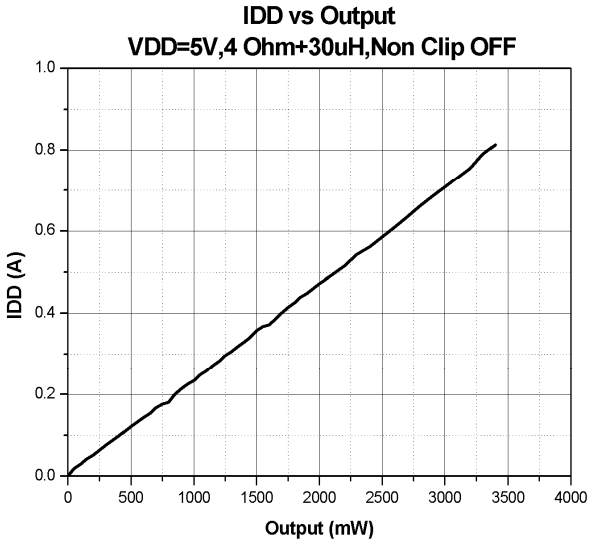


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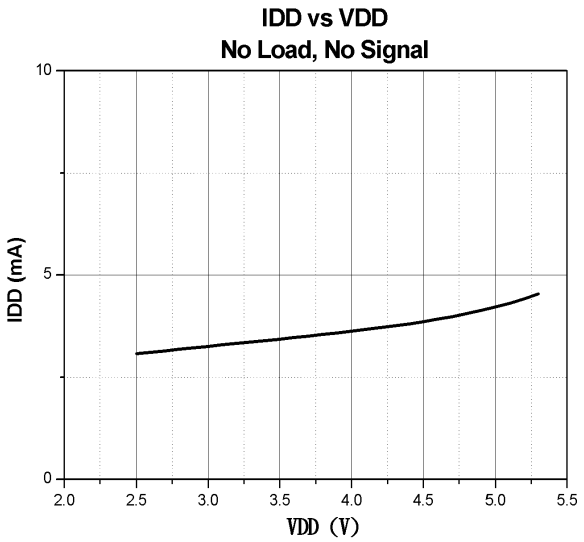


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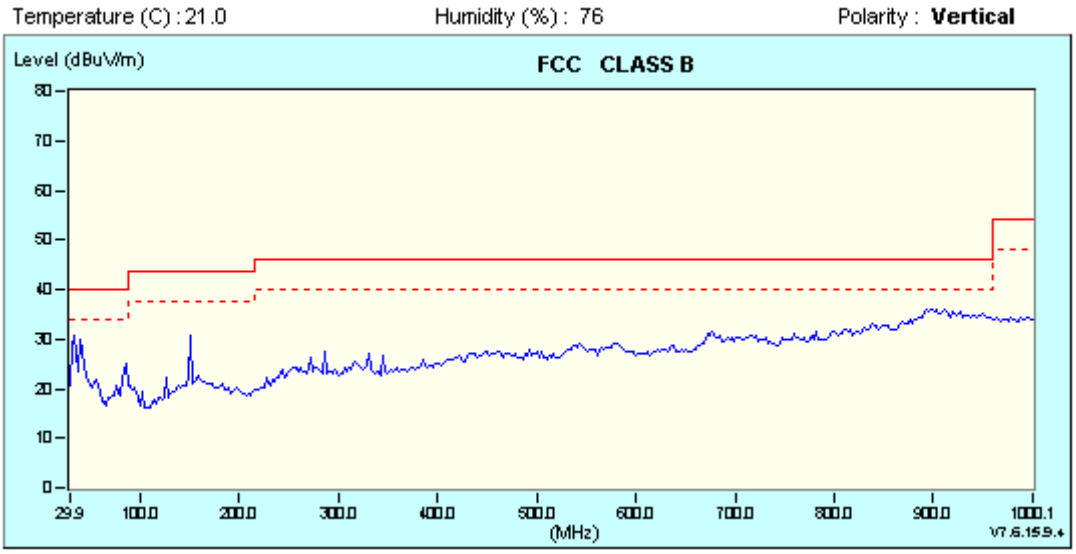
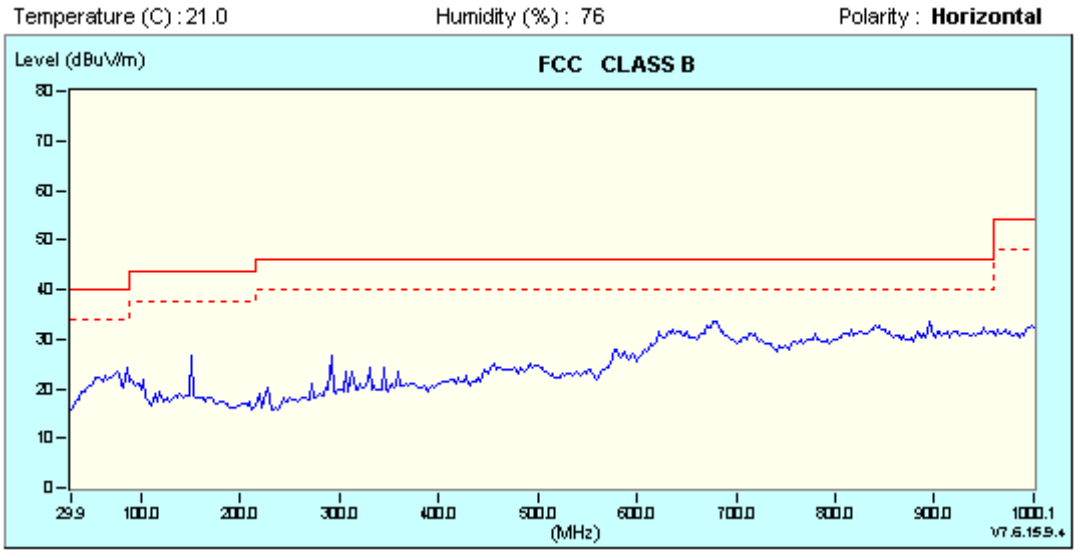


Figure26.



Note: The result of EMI measurement using EUTECH's evaluation board. However, this does not guarantee the effectiveness in your product with this device implemented. For actual use, please evaluate the effectiveness, according to your use conditions.

Figure27.

Application Information

Fully Differential Amplifier

The EUA2068A is a fully differential amplifier that features differential inputs and outputs. The EUA2068A also includes a common mode feedback loop that controls the output bias value to average it at $V_{DD}/2$ for any DC common mode input voltage. This allows the device to always have a maximum output voltage swing, and by consequence, maximize the output power. Moreover, as the load is connected differentially, compared to a single-ended topology, the output is four times higher for the same power supply voltage. The fully differential EUA2068A can still be used with a single-ended input; however, the EUA2068A should be used with differential inputs when in a noisy environment, like a wireless handset, to ensure maximum noise rejection.

Advantages of Fully Differential Amplifiers

The advantages of a fully-differential amplifier are:

- Very high PSRR (Power Supply Rejection Ratio).
- High common mode noise rejection.
- Virtually zero pop without additional circuitry, giving an faster start-up time compared to conventional single-ended input amplifiers.
- No input coupling capacitors required thanks to common mode feedback loop.
- Midsupply bypass capacitor not required.

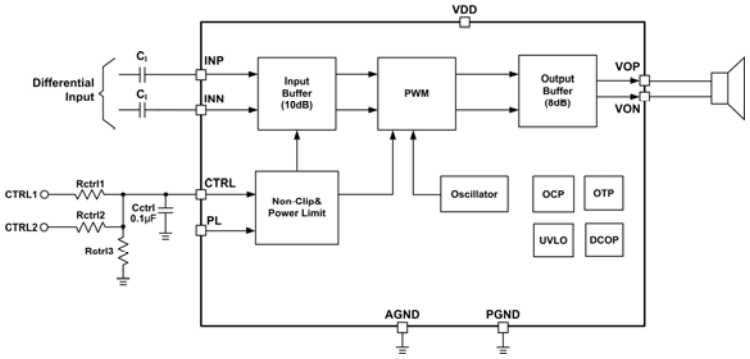


Figure 28. Differential Input Configuration and Input Capacitors

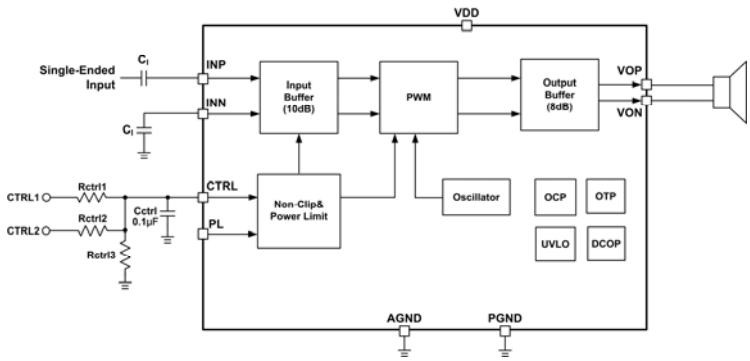


Figure 29. Single-Ended Input Configuration

Input Signal Capacitor

An analog signal, ranging from GND to VDD, should be input to input pins (INP/INN) through the DC-cut capacitor C_{IN}.

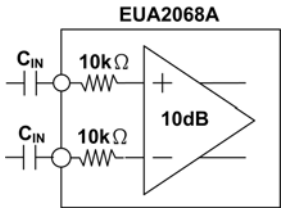


Figure 30.

The DC-cut capacitor C_{IN} and the input impedance 10kΩ make a high-pass filter, the cut-off frequency

$$fc = \frac{1}{2\pi \times C_{IN} \times (10k\Omega)}$$

NCPL (Power Limit Interlock Non-Clip) Function

NCPL function allows Non-Clip function to be enabled even though the power limit setting is performed. Non-Clip function automatically controls the PWM amplifier gain so as not to clip the output signals even if the speaker output is clipped at the supply voltage. This control, following the supply-voltage fluctuation, has an effect also on the case that a supply voltage fluctuates, such as a battery etc. And, Power Limit function, allowing a limiting level (Max. Output) to be arbitrarily set, is available for speaker overload protection etc.

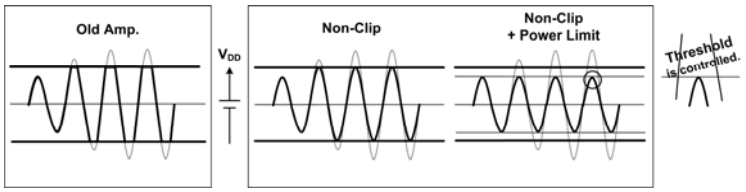


Figure 31. NCPL (Power Limit Interlock Non-Clip) Conceptual diagram

- Non-Clip Function

Previously, when using a device in an application where the supply voltage may fluctuate, the source volume had been set so as not to distort the sound even at the minimum voltage.

But, Non-Clip function allows source volume to be set based on the maximum voltage without regard for distortion. This function automatically controls the gain according to the variation for a supply voltage; therefore, this is an optimum function for an application that needs a sound pressure level to some extent, such as a case using a small speaker, because the sound pressure is increased.

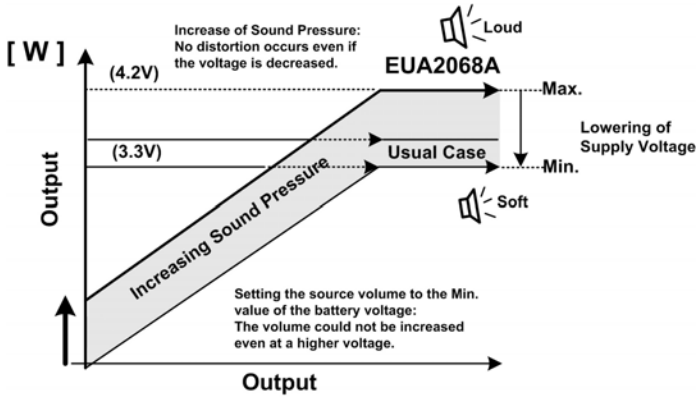


Figure 32. Increase of Sound Pressure using Non-Clip Function-Conceptual Diagram

● Power Limit Function

Power Limit function is a function provided for the purpose of reducing the overload on a speaker and limits the output power *1. Any power limit level, ranging from 100mW to 1000mW, can be freely set. Even if an excessive signal is input, Non-Clip function automatically controls PWM amplifier gain to keep the maximum output power to a constant level, And, when PL pin is connected to GND, the output is set to 0.60W (R_L=8Ω).

*1:This function does not guarantee 100% speaker protection.

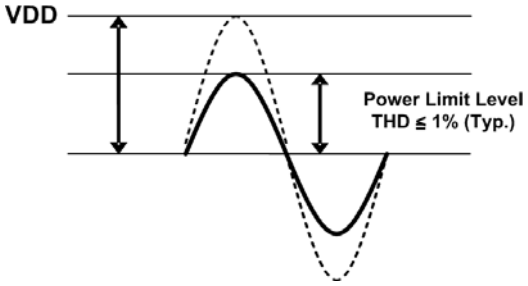


Figure 33. Power Limit-Conceptual Diagram

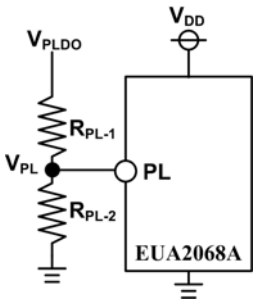


Figure 34. An example of setting PL pin

The voltage at V_{PL} is calculated by the following formula:

$$V_{PL} = \frac{R_{PL-2} \times V_{PLDO}}{R_{PL-1} + R_{PL-2}}$$

Table 1. A typical example of setting PL Pin

V_{PLDO}	R_{PL-1}	R_{PL-2}	V_{PL}	Power Limit Output (R_L=8Ω)
1.8V	100kΩ	240kΩ	1.27V	700 mW
	130kΩ	270kΩ	1.22V	650 mW
Fix Mode			0V	600 mW
1.8V	110kΩ	180kΩ	1.12V	550 mW
	110kΩ	160kΩ	1.07V	500 mW
	100kΩ	130kΩ	1.02V	450 mW
	160kΩ	180kΩ	0.95V	400 mW

Protection Function

EUA2068A has the following protection functions for the digital amplifier; DC Output Protection Function, Over-current Protection function, Thermal Protection function, and Low voltage Malfunction Prevention function.

● DC Output Protection Function

This function protects the speaker by detecting a DC output condition and stopping the output. This is because DC output may cause abnormal heat in the speaker's voice coil if a DC voltage keeps on applying to the speaker.

The protection function works when an output voltage in excess of the DC error detection voltage V_{DCDET} keeps on outputting for more than the DC error detection time (T_{DCDET}).

This protection function is provided not for maintaining the speaker protection but for ensuring safety, and does not guarantee the protection of the speaker.

● Over-current Protection Function

This is the function to establish the over-current protection mode when detecting a short circuit between EUA2068A differential output terminal and VSS, VDD, or another differential output. When a short circuit event happens, the EUA2068A stops the outputs and tries to reactivate itself after few milliseconds. This auto-recovery will continue until the short circuit event is removed.

● Thermal Protection Function

This is the function to establish the thermal protection mode when detecting excessive high temperature of EUA2068A itself. In the thermal protection mode, the differential output terminal becomes Hi-Z state. And, when EUA2068A gets out of such condition, the protection mode is cancelled.

● Low voltage Malfunction Prevention function

This is the function to establish the low voltage protection mode when VDD terminal voltage becomes lower than the detection voltage (V_{UVLL}) for the low voltage malfunction prevention and to cancel the protection mode when VDD terminal voltage becomes higher than the threshold voltage (V_{UVLH}) and by return procedure form shut down for its deactivation.

(In sag state , this function works and EUA2068A becomes a low voltage protection mode.)

In the low voltage protection mode, the differential output pin becomes Weak Low state (a state grounded through high resistivity). EUA2068A will start up within the start-up time (T_{STUP}) when the low voltage protection mode is cancelled.

Control Function

● Shut down and Initialization function

When CTRL terminal is connected to GND potential, the IC goes to the shut-down mode. In the mode, all the circuit functions stop and its current consumption becomes the lowest. And, the output terminals become Weak Low (A high resistance grounded state).

When in the shut-down mode, the level of the terminal must not be changed from GND level during t_{PDWL}.

On the contrary, when CTRL terminal is set to H level, the shut-down mode is canceled and the IC starts up after startup time (t_{STUP}).

In order to return from the shut-down mode a desired mode needs to be set after setting both CTRL1 and CTRL2 to H level during t_{PDWH}. In addition, at startup, cancel the shut-down mode after supply voltages has been sufficiently stabilized.

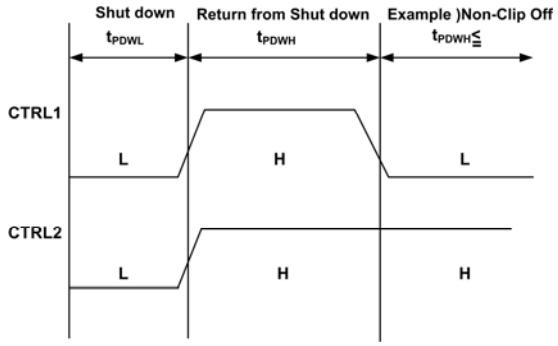


Figure 35.

● CTRL terminal function

By connection external resistors (Rctrl1, Rctrl2, and Rctrl3: Accuracy of 1%) to CTRL terminal, and impression setting threshold voltage of each mode to CTRL terminal, the followings can be set: Non-Clip A, Non-clip B, Non-Clip OFF, and shut-down mode. When turning on the supply voltage or canceling the shut-down mode, control the CTRL terminal according to procedure for canceling shut-down (See Page6.)

Connect the terminal to the ground through a capacitor Cctrl (a ceramic capacitor of 0.1μF or more).

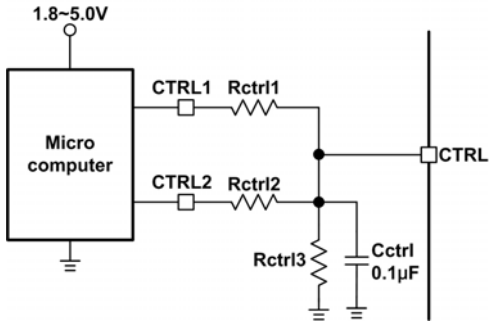


Figure 36.

Table 2.

CTRL1	CTRL2	Function
H	H	Non-Clip A mode
H	GND	Non-Clip B mode
GND	H	Non-Clip Off mode
GND	GND	Shut-down mode

"H" level indicates a microcomputer's I/O port H level output voltage that is input to CTRL1 and CTRL2 terminals and GND indicates GND of the microcomputer.

GND level of the microcomputer must be the same as that of EUA2068A.

The control of CTRL terminal is based on I/O port H level output voltage of microcomputer that is connected.

Set resistance constants according to I/O port H level output voltage of each microcomputer as shown below.

Table 3.

I/O port H level output voltage of microcomputer	1.8V	2.6V	3.0V	3.3V	5.0V
Rctrl 1	2.2kΩ	8.2kΩ	8.2kΩ	8.2kΩ	8.2kΩ
Rctrl 2	3.3kΩ	12kΩ	12kΩ	12kΩ	12kΩ
Rctrl 3	68kΩ	12kΩ	8.2kΩ	5.6kΩ	2.7kΩ

Functions of CTRL pin are designed with their control by two control pins (CTRL1 and CTRL2).

Only a switching control between Non-Clip A mode and Shut-down mode is available when a single control terminal is used. A setting voltage should be set according to V_{MOD1} AND V_{MOD4}, and use a RC filter with time constant of 1msec or more in order to eliminate noise at transmission side such as Micon etc. (Example. Rctrl=10kΩ and Cctrl=0.1μF).

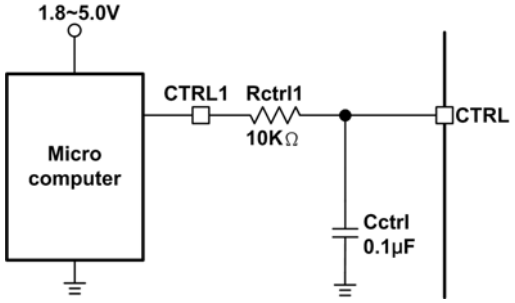


Figure 37.

Table 4.

CTRL1	Function
H	Non-Clip A mode
GND	Shut-down mode

Quick Start/Quick Mute Function

Pop noise reduction circuit (Quick Start/ Quick Mute) operates at startup or shutdown of the speaker amplifier.

This is a function to vary ON/OFF envelopes at a slow rate. Even a high-speed ON/OFF operation, this reduces intermittent sound considerably and eliminates uncomfortable feeling.

That is, Quick Start/ Quick Mute is a pop noise reduction function with Non-Clip technique applied.

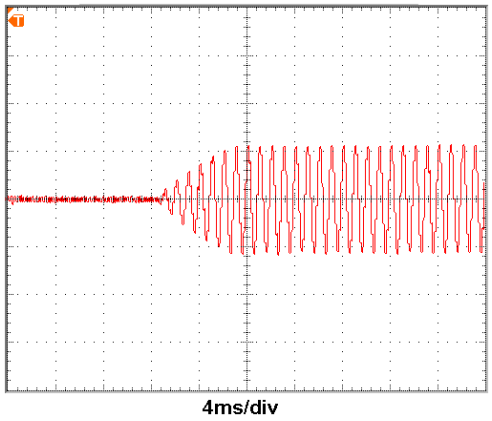


Figure 38. Quick Start

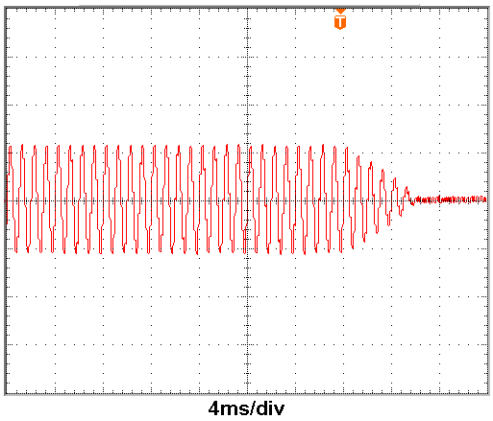


Figure 39. Quick Mute

Power Supply Decoupling Capacitor (C_S)

The EUA2068A is a high-performance CMOS class-D audio amplifier that requires adequate power supply decoupling to ensure the efficiency is high and total harmonic distortion (THD) is low. For higher frequency transients, spikes, or digital hash on the line, a good low equivalent-series-resistance (ESR) ceramic capacitor, typically 1μF, placed as close as possible to the device V_{DD} lead works best. Placing this decoupling capacitor close to the EUA2068A is very important for the efficiency of the class-D amplifier, because any resistance or inductance in the trace between the device and the capacitor can cause a loss in efficiency. For filtering lower-frequency noise signals, a 10μF or greater capacitor placed near the audio power amplifier would also help.

PCB Layout

As output power increases, interconnect resistance (PCB traces and wires) between the amplifier, load and power supply create a voltage drop. The voltage loss on the traces between the EUA2068A and the load results is lower output power and decreased efficiency. Higher trace resistance between the supply and the EUA2068A has the same effect as a poorly regulated supply, increase ripple on the supply line also reducing the peak output power. The effects of residual trace resistance increases as output current increases due to higher output power, decreased load impedance or both. To maintain the highest output voltage swing and corresponding peak output power, the PCB traces that connect the output pins to the load and the supply pins to the power supply should be as wide as possible to minimize trace resistance.

The use of power and ground planes will give the best THD+N performance. While reducing trace resistance, the use of power planes also creates parasite capacitors that help to filter the power supply line.

The inductive nature of the transducer load can also result in overshoot on one or both edges, clamped by the parasitic diodes to GND and V_{DD} in each case. From an EMI stand- point, this is an aggressive waveform that can radiate or conduct to other components in the system and cause interference. It is essential to keep the power and output traces short and well shielded if possible. Use of ground planes, beads, and micro-strip layout techniques are all useful in preventing unwanted interference.

As the distance from the EUA2068A and the speaker increase, the amount of EMI radiation will increase since the output wires or traces acting as antenna become more efficient with length. What is acceptable EMI is highly application specific.

Ferrite bead and capacitance placed close to the EUA2068A may be needed to reduce EMI radiation. Select a ferrite bead with the high impedance around 100MHz and a very low DCR value in the audio frequency range is the best choice. The MPZ1608S221A1 from TDK is a good choice.

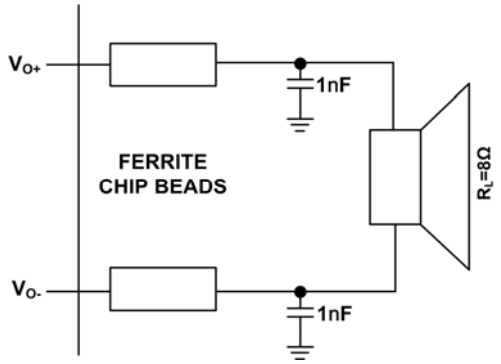
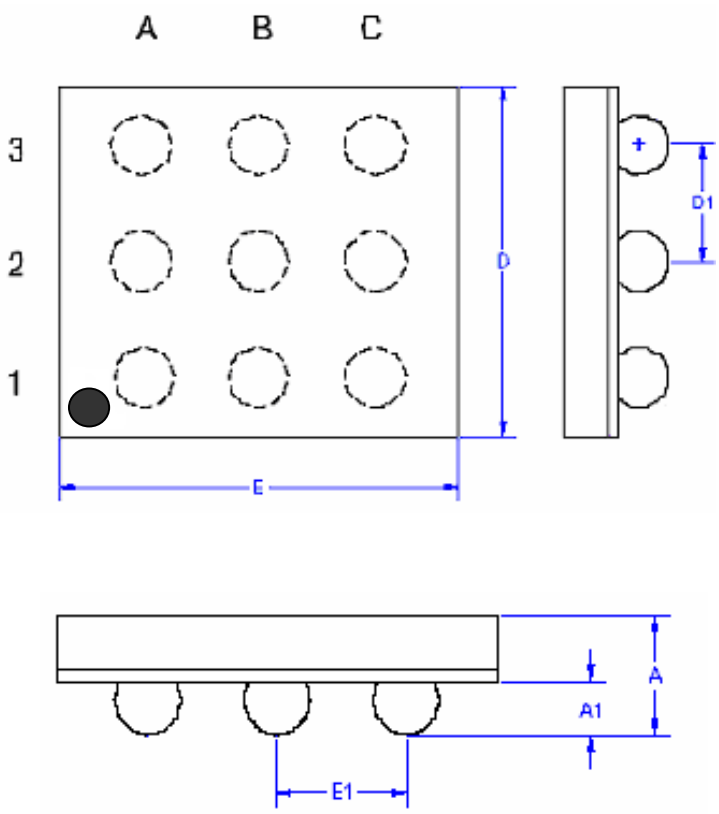


Figure 40. Optional EMI Ferrite Bead Filter

Packaging Information

WCSP-9



SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	-	0.675	-	0.027
A1	0.15	0.35	0.006	0.014
D	1.45	1.55	0.057	0.061
D1	0.50		0.020	
E	1.45	1.55	0.057	0.061
E1	0.50		0.020	